

SPENCER FOLK

spencer.folk@gmail.com

spencerfolk.github.io

EDUCATION

2019-present	University of Pennsylvania , <i>Ph.D. Candidate, Mechanical Engineering</i> Advisors: Dr. Vijay Kumar , Dr. Mark Yim	Philadelphia, PA
2019-2022	University of Pennsylvania , <i>M.S., Robotics</i>	Philadelphia, PA
2015-2019	Lafayette College , <i>B.S., Mechanical Engineering</i> Graduated Summa Cum Laude	Easton, PA

SKILLS

Programming Languages	Python, C++, MATLAB
Software Frameworks and Tools	ROS, Docker, PyTorch, Git, GitHub Actions
Design and Fabrication	SOLIDWORKS, Laser Cutting, 3D Printing, Soldering
Communication	LaTeX, Microsoft Office, Adobe Premiere Pro

RESEARCH EXPERIENCE

2022-present	Wind Estimation and Motion Planning for Urban Air Mobility <i>Ames Research Center, National Aeronautics and Space Administration (NASA)</i>	Mountain View, CA
– Devised a deep learning-based framework for predicting local urban wind flow fields from on-board range and wind measurements, enabling real-time wind flow field predictions on computationally constrained aerial vehicles. – Developed and deployed a GPU-accelerated receding horizon optimal controller on flight-ready hardware, leveraging local wind predictions from neural networks to reduce UAV energy consumption and crash rates by up to 30% in windy urban environments. – Implemented fast computational fluid dynamics code on a high-performance computing cluster to rapidly generate massive datasets of simulated urban winds for applications in deep learning. – Conducted the first set of hardware experiments in a novel open-air wind tunnel facility, architecting the end-to-end software stack enabling autonomous flight testing.		
2019-present	Graduate Research on Aerial Robotics <i>GRASP Lab, University of Pennsylvania</i>	Philadelphia, PA
– Investigating state-of-the-art methods in deep learning, state estimation, and robot motion planning to enhance situational awareness of multirotor UAVs in cluttered windy environments to fly safer and more efficiently. – Building and maintaining a popular aerial robotics simulator, RotorPy, providing an entry point for learning motion planning, state estimation, and control algorithms for aerial robots. – Authoring and submitting manuscripts to top-tier robotics and learning venues including IEEE RA-L, IEEE ICRA, CoRL, and ASME JMR. – Managing teams of undergraduate and graduate students supporting our projects.		
2019	Undergraduate Research on Ethorobotics <i>Department of Mechanical Engineering, Lafayette College</i>	Easton, PA
– Modeled the movement of Archerfish in schooling and hunting sequences with Markov Decision Processes and stochastic differential equations.		

- Replicated fish swimming on a 5-DoF robotic fish to successfully elicit specific social behaviors among Archerfish, culminating in a journal paper in *Biological Cybernetics*.

- 2017-2019 Design and Fabrication of 3D-Printed UAVs Aberdeen, MD**
United States Army Research Laboratory
- Designed, manufactured, and assembled parameter-driven 3D-printed quadrotor UAVs that reduced part and fastener count by 58%, satisfying Army requirements for a mission-specific UAV available on demand within 24 hours of request.
 - Developed a routine using textbook rotorcraft mechanics that appropriately sized and outfitted a quadrotor to optimize for mission parameters like range, endurance, loiter time, and payload capacity.
 - Applied machine learning to identify max endurance and max range flight speeds for 3D-printed UAVs based on flight data, capturing often unmodeled effects like profile drag, rotor-rotor interference, and rotor-frame interference.

HIGHLIGHTED PUBLICATIONS

Folk, S., Melton, J., Margolis, B. W. L., Yim, M., & Kumar, V. (2025). Towards Safe and Energy-Efficient Real-Time Motion Planning in Windy Urban Environments. *International Conference on Robotics and Automation (ICRA) 2025*. [Conference]

Folk, S., Melton, J., Margolis, B. W. L., Yim, M., & Kumar, V. (2024). Learning Local Urban Wind Flow Fields from Range Sensing. *IEEE Robotics and Automation Letters*. [Journal]

Sanghvi, H., **Folk, S.,** & Taylor, C. J. (2024). OCCAM: Online Continuous Controller Adaptation with Meta-Learned Models. *Conference on Robot Learning (CoRL)*. [Conference]

Zhang, H., Srikanthan, A., **Folk, S.,** Kumar, V., & Matni, N. (2024). Why change your controller when you can change your planner: Drag-aware trajectory generation for quadrotor systems. *arXiv preprint arXiv:2401.04960*. [Preprint]

Weakly, J., Li, X., Agarwal, T., Li, M., **Folk, S.,** Jiang, C., & Sung, C. (2024). Bistable Aerial Transformer: A Quadrotor Fixed-Wing Hybrid That Morphs Dynamically Via Passive Soft Mechanism. *Journal of Mechanisms and Robotics*, 16(7), 071016. [Journal]

Folk, S., Paulos, J., & Kumar, V. (2023). RotorPy: A Python-based Multirotor Simulator with Aerodynamics for Education and Research. *The Role of Robotic Simulators for Unmanned Aerial Vehicles – ICRA 2023*. [Conference Workshop]

Brown, A. A., Brown, M. F., **Folk, S. R.,** & Utter, B. A. (2021). Archerfish respond to a hunting robotic conspecific. *Biological Cybernetics*, 115(6), 585-598. [Journal]

INVITED TALKS

“Towards Safe and Efficient Real-Time Motion Planning in Windy Urban Environments”

East Coast Meetup, Dronecode Foundation, Philadelphia, PA.

MEAM Departmental Seminar Series, University of Pennsylvania, Philadelphia, PA.

Nikolai Matni Group, University of Pennsylvania, Philadelphia, PA.

March 2025

February 2025

October 2024

“Learning-Based Methods for Real-Time Wind Prediction in Urban Environments”

Intelligent Robot Motion Lab x FAST Group, Princeton University, Princeton, NJ.

May 2024

LEADERSHIP AND MENTORING

- 2020-present** **Virtual High School Mentoring**
Polygence
- Fostering curiosity and interest in STEM among high school students by guiding them through self-motivated projects, which have included designing a gimbaled rocket, solar-powered UAV, and an aerial landmine detection system.
- 2020-2022** **Teaching Assistantships**
University of Pennsylvania
- Graduate courses: Advanced Robotics (600 level).
 - Undergraduate courses: Control for Autonomous Robots (400 level), Vibrations (300 level), Statics and Strengths of Materials (200 level).
- 2021** **Vice President, Mechanical Engineering Graduate Association**
University of Pennsylvania
- Upheld a sense of community through the COVID-19 pandemic by organizing virtual events ranging from Zoom trivia nights to wellness challenges.
- 2018-2019** **Chapter President, Tau Beta Pi (PA-E)**
Lafayette College
- Organized chapter and community events, communicated with headquarters to maintain national standards, and represented the chapter at national and district conferences.

HONORS AND AWARDS

- 2019** **Graduate Assistance in Areas of National Need (GAANN) Fellowship**
University of Pennsylvania
Awarded to “graduate students of superior ability planning to pursue the highest degree in their course of study.”
- 2019** **Karl J. Ammerman Prize**
Lafayette College
Awarded to the “most deserving student in the mechanical engineering department, as selected by the faculty of the department.”
-